

SHORT-CYCLE COUNTEREXAMPLES TO CONJECTURE 12 OF BERMAN–GÉVAY–PISANSKI

ABSTRACT. Berman, Gévay, and Pisanski proposed two parameterized cyclic-voltage families for geometric $(7m)_4$ configurations, while noting that further parameter restrictions might be necessary. We determine exactly when the corresponding derived multigraphs avoid cycles of length two and four. This refutes the sufficiency of the parameter ranges displayed in their Conjecture 12: the derived graph of $F_2(m; 2, 1)$ has a 4-cycle for every $m \geq 5$, and that of $F_1(m; m/2, b)$ has a 4-cycle for every even $m \geq 4$ and every $1 \leq b \leq m/2$. Hence these assignments do not yield even combinatorial configurations. We do not address geometric realizability of the surviving lifts.

1. THE VOLTAGE FAMILIES

For $m > 3$, Berman, Gévay, and Pisanski introduced a generic reduced Levi graph $H = \text{RLG}(B)$ and, in Conjecture 12 of [1] (Conjecture 7.1 of the preprint version listed there, whose statement is identical), proposed two families of cyclic voltage assignments over $\mathbb{Z}_m$, in the parameter order

$$(a, c, d, e, f, g, q, a', c', d', e', f', g', q', t) :$$

$$F_1(m; a, b) = (a, -a, a, b, b, -a, -2a, a, -a, a, b, b, -a, -2a, 0),$$

$$F_2(m; a, b) = (a, b, b, b, b, b, b, a, b, b, b, b, b, b, 0).$$

The displayed range for F_1 is

$$a \geq b, \quad 1 \leq a, b \leq m/2,$$

and that for F_2 is

$$a \geq b, \quad a \neq m/2.$$

The conjecture is stated with the proviso that “there may be other constraints on a and b that have not yet been identified” [1]. Our purpose is to determine exactly the constraints forced by cycles of length at most four.

The line-orbit vertices of H are

$$M, R, P, B, C, Y, G,$$

and the point-orbit vertices are

$$r, m, p, y, g, c, b.$$

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This notation is inherited from [1] and overloads four letters: b , c and g name both point orbits and voltages, and m is both the modulus and a point orbit. Subscripted lower-case letters always denote lifted points, so m_1 in the proof of Corollary 2 is a point of the derived graph and not the modulus.

Using the standard cyclic-voltage formalism for polycyclic configurations [2], and following [1, Sec. 2], an edge of voltage $\alpha \in \mathbb{Z}_m$ means that the lifted line L_i is incident with the lifted point $v_{i+\alpha}$. Transcribing the generic assignment drawn in Figure 4(b) of [1] into incidence form gives

$$(1) \quad \begin{aligned} M &: r(0), r(a), y(c), p(q), & R &: m(0), m(a'), g(d), p(0), \\ P &: r(0), y(0), g(0), m(q'), & B &: y(e'), g(0), c(g), c(t), \\ C &: y(0), g(f'), b(0), b(g'), & Y &: m(c'), b(e), c(0), p(0), \\ G &: r(d'), b(0), c(f), p(0). \end{aligned}$$

Provenance of (1). Figure 4(b) of [1] is a drawing, so (1) is a transcription of it rather than a quotation, and everything below is a statement about the graph that (1) specifies. We therefore print the transcription in full and record the checks that pin it down. Of the two checks that follow, the structural one is internal to (1) and can be carried out from the displayed data alone; the numerical one is the single point at which this note appeals to data outside it, namely the incidence table of [1, Table 1], which we do not reproduce here.

Structurally, (1) lists 28 edges on the 14 vertices of H , four at each vertex; the 13 edges of voltage 0 form a spanning tree of H , and the 15 named voltages occupy the 15 remaining edges, one each. This is the normalization realized by the fifteen edge labels $a, c, d, e, f, g, q, a', c', d', e', f', g', q', t$ carried by the figure.

Numerically, substituting $F_1(3; 1, 1)$, that is

$$(a, c, d, e, f, g, q, a', c', d', e', f', g', q', t) \\ = (1, 2, 1, 1, 1, 2, 1, 1, 2, 1, 1, 2, 1, 1, 2, 1, 0) \quad \text{in } \mathbb{Z}_3,$$

into (1) reproduces all 28 incidences of the incidence table of $B(21_4)$ in [1, Table 1] after the relabelling $i \mapsto -i$ of $\mathbb{Z}_3$, and [1] record that $F_1(3; 1, 1)$ is isomorphic to $B(21_4)$. The relabelling is harmless, because $i \mapsto -i$ is an automorphism of the derived graph and so preserves girth. In particular the four parallel pairs of (1), namely $Mr = \{0, a\}$, $Rm = \{0, a'\}$, $Bc = \{g, t\}$ and $Cb = \{0, g'\}$, then carry the voltage differences 1, 1, 2, 2, matching the four points of that table which lie on two lines of a single orbit: r_0 on M_0, M_1 ; m_0 on R_0, R_1 ; c_0 on B_0, B_2 ; b_0 on C_0, C_2 .

At $m = 3$ with $a = b = 1$ several of the fifteen voltages take equal values, so this comparison alone cannot detect a permutation of labels within a block of equal values. The degeneracy is genuine: at $F_1(3; 1, 1)$ the generic values a , b and $-2a$ all equal 1 in $\mathbb{Z}_3$, so the comparison of parallel pairs

shows only that the label carried by Mr has generic value a , b or $-2a$, and not that it is a itself. We therefore do not claim that (1) is the only reading of Figure 4(b) compatible with these checks, and we claim nothing beyond it: Theorem 1 and Corollary 2 are assertions about the graph that (1) specifies. Consequently a reader without [1, Table 1] at hand can still check everything in the sections that follow, taking (1) as the definition of H and of its voltages; what rests on the comparison with that table is only the identification of this graph with the one drawn in Figure 4(b) of [1], and hence the bearing of our results on Conjecture 12 as its authors intended it. The four parallel pairs and the ten quadrilaterals used in the proof below are features of H itself, so a reader who reads the figure differently need only substitute that reading into (1) and repeat the proof with the labels permuted accordingly.

Let $\Gamma_i(m; a, b)$ denote the derived multigraph associated with $F_i(m; a, b)$. A Levi graph of a combinatorial point-line configuration is simple and has no 4-cycle; thus the relevant derived graph must have girth at least six, where a parallel pair is regarded as a cycle of length two.

2. SHORT-CYCLE CLASSIFICATION

Theorem 1. *Let $m \geq 2$ and let $a, b \in \mathbb{Z}_m$.*

(1) *The graph $\Gamma_1(m; a, b)$ has girth at least six if and only if*

$$a \neq 0, \quad b \neq 0, \quad 2a \neq 0, \quad 2b \neq 0 \quad \text{in } \mathbb{Z}_m.$$

(2) *The graph $\Gamma_2(m; a, b)$ has girth at least six if and only if*

$$a, b, 2a, 2b, a - b, a - 2b \neq 0 \quad \text{in } \mathbb{Z}_m.$$

Proof. A cycle in a derived voltage graph projects to a cyclically reduced closed walk of the same length in the base graph, and the net voltage of that walk is zero. Conversely, a cyclically reduced zero-voltage closed walk lifts to a nonbacktracking closed walk and therefore contains a cycle of no greater length. Since the derived graphs are bipartite, it is enough to enumerate the projected closed walks of lengths two and four.

The four parallel pairs in H are Mr , Rm , Bc , and Cb . Their voltage differences, up to sign, are

$$a, \quad a', \quad g - t, \quad g',$$

and traversing a parallel pair twice gives

$$2a, \quad 2a', \quad 2(g - t), \quad 2g'.$$

The four parallel pairs are pairwise vertex-disjoint, so no vertex of H lies in two of them. A cyclically reduced closed walk of length four visits four, three, or two distinct vertices. On three vertices it has the form $L - v_1 - L - v_2 - L$ or $L_1 - v - L_2 - v - L_1$; the first requires the two pairs Lv_1 and Lv_2 to be parallel and the second requires vL_1 and vL_2 to be parallel, so each would place the repeated vertex, L respectively v , in two distinct parallel pairs, and

both are excluded. On two vertices it traverses a single parallel pair twice, which is the case already treated. Hence every remaining cyclically reduced closed walk of length four visits four distinct base vertices and traces one of the ten quadrilaterals below. Directly from (1), their net voltages are as follows, up to sign:

walk	net voltage	walk	net voltage
$MrPyM$	$c, a - c$	$RmYpR$	$c', a' - c'$
$GbYpG$	e	$GcYpG$	f
$GbYcG$	$e + f$	$ByPgB$	e'
$CyPgC$	f'	$ByCgB$	$e' + f'$
$GrMpG$	$d' + q, a - d' - q$	$PgRmP$	$d + q', a' - d - q'$

These four parallel pairs and ten quadrilaterals exhaust the cyclically reduced closed walks of lengths two and four in H .

Substitution of F_1 leaves, up to sign, precisely

$$a, \quad b, \quad 2a, \quad 2b.$$

Substitution of F_2 leaves, up to sign, precisely

$$a, \quad b, \quad 2a, \quad 2b, \quad a - b, \quad a - 2b.$$

The two assertions follow. $\square$

Corollary 2. *The displayed parameter ranges in Conjecture 12 of [1] are not sufficient.*

- (1) *For every $m \geq 5$, the assignment $F_2(m; 2, 1)$ satisfies the displayed conditions, and even*

$$1 \leq 1 < 2 < m/2,$$

but its derived graph has a 4-cycle. In particular,

$$F_2(5; 2, 1) = (2, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 0)$$

does not yield a (35_4) configuration, geometric or combinatorial.

- (2) *For every even $m \geq 4$ and every $1 \leq b \leq m/2$, the assignment $F_1(m; m/2, b)$ satisfies the displayed conditions but its derived graph has a 4-cycle.*

Proof. For $F_2(m; 2, 1)$, equation (1) gives

$$P_0 \sim g_0, m_1, \quad R_{-1} \sim g_0, m_1.$$

The vertices P_0, R_{-1}, g_0, m_1 are distinct, so

$$P_0 - g_0 - R_{-1} - m_1 - P_0$$

is a 4-cycle. Equivalently, the obstruction $a - 2b$ in Theorem 1 vanishes. For $m \geq 5$, the choice $(a, b) = (2, 1)$ satisfies the displayed conditions of Conjecture 12 and also the stated strict half-range.

Now let m be even and set $a = m/2$. The two lifted lines M_0 and M_a both contain the two distinct points r_0 and r_a , because $2a = 0$ in $\mathbb{Z}_m$. Hence

$$M_0 - r_0 - M_a - r_a - M_0$$

is a 4-cycle. A point-line configuration cannot have two distinct lines through two distinct common points, so none of these assignments yields the asserted configuration. $\square$

Remark 3. Conjecture 12 explicitly allows that further restrictions may be necessary. Accordingly, the result above corrects the displayed sufficiency ranges; it does not settle geometric realizability for the remaining parameter values.

Since $\Gamma_i(m; a, b)$ is bipartite and 4-regular with parts of size $7m$, girth at least six is not merely necessary but equivalent to the lift being a combinatorial $(7m)_4$ configuration. Theorem 1 therefore determines exactly which of these assignments yield combinatorial configurations, and geometric realizability is the only question it leaves open.

Within the printed range for F_1 , Theorem 1 says that the short-cycle condition is exactly $a < m/2$. The printed range for F_2 places no upper bound on a and does not separate a from b , so besides $a = 2b$ it also admits $a = b$, for instance $F_2(m; 1, 1)$ for every $m > 3$, and $b = m/2$ with $a > m/2$, for instance $F_2(6; 4, 3)$; these make $a - b$ and $2b$ vanish respectively. The F_2 range is thus insufficient for every $m > 3$, not only for $m \geq 5$. Within the stronger range $1 \leq b < a < m/2$ it is exactly $a \neq 2b$.

None of this contradicts an assignment displayed in [1]. The illustrated examples $F_1(4; 1, 1)$, $F_1(5; 2, 2)$, $F_1(6; 2, 1)$ and $F_2(4; 3, 1)$, $F_2(5; 3, 1)$, $F_2(6; 4, 1)$ of [1, Figs. 11 and 12] all satisfy the criteria of Theorem 1: in each of the three F_1 examples none of a , b , $2a$, $2b$ vanishes, and in each of the three F_2 examples $a - b$ and $a - 2b$ do not vanish either. The two lists of criteria differ, and the F_1 example $F_1(6; 2, 1)$ does have $a = 2b$; by Theorem 1 that is an obstruction for F_2 and not for F_1 . The counterexamples above occupy parameter values that lie inside the printed ranges but were not illustrated. That $F_2(4; 3, 1)$ has $a > m/2$ also confirms that the bound $1 \leq a, b \leq m/2$ is imposed on F_1 alone. We are not aware of prior work identifying the missing constraints.

REFERENCES

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